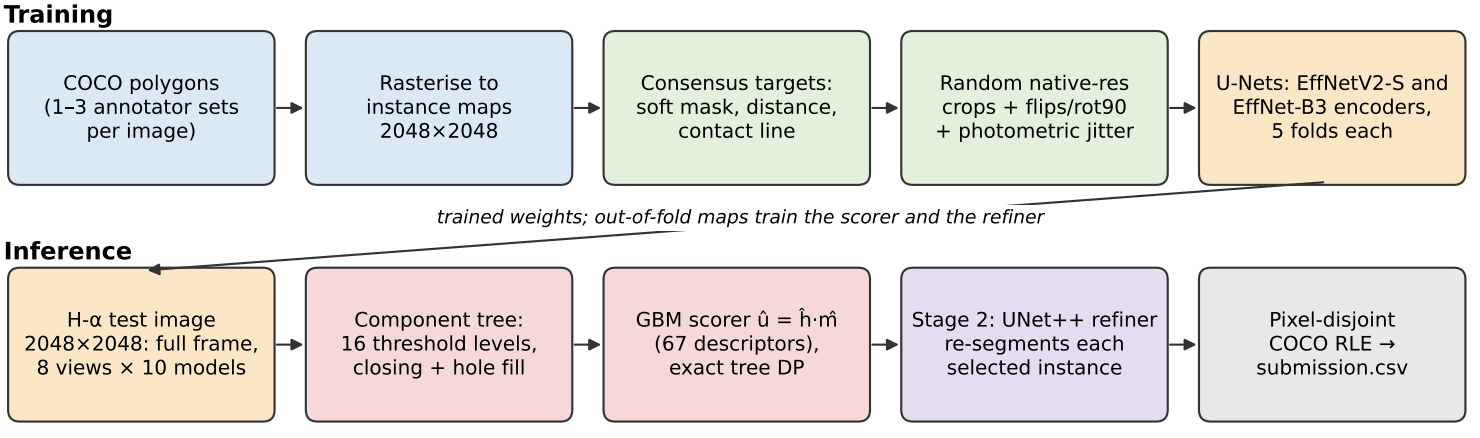
Training
COCO polygons (1–3 annotator sets per image)
Rasterise to instance maps 2048×2048
Consensus targets: soft mask, distance, contact line
Random native-res crops + flips/rot90 + photometric jitter
U-Nets: EffNetV2-S and EffNet-B3 encoders, 5 folds each
trained weights; out-of-fold maps train the scorer and the refiner
Inference
H-α test image 2048×2048: full frame, 8 views × 10 models
Component tree: 16 threshold levels, closing + hole fill
GBM scorer û = ĥ·m̂ (67 descriptors), exact tree DP
Stage 2: UNet++ refiner re-segments each selected instance
Pixel-disjoint COCO RLE → submission.csv